

Business Problem Statement

A leading retail company aims to decode customer shopping behavior to maximize sales, elevate customer satisfaction, and foster long-term brand loyalty. Management has observed shifts in purchasing trends across geographic locations, preferred payment methods, seasonal category demands, and discount responses.

To address these challenges, the leadership team requires a deep-dive analysis into critical operational areas:

- Geographic market performance and core customer age demographics
- Transactional preferences across payment methods
- Seasonal category performance for targeted inventory planning
- The psychological impact of discounts on product review ratings and customer feedback
- Demographic revenue contributions (gender and age groups)
- Subscription program effectiveness and high-spend behavior
- Customer lifecycle segmentation (New, Returning, and Loyal buyers)
- Top-performing product tiers within each category

You are tasked with analyzing the company's consumer behavior dataset to answer the following overarching business question:

"How can the company leverage consumer shopping data to identify geographic, seasonal, and behavioral trends, improve customer engagement, and optimize marketing and product strategies?"

Project Deliverables

1. **Data Preparation & Modeling (Python):** Clean, impute missing values, transform features, and format the raw dataset for database storage.
2. **Data Analysis (SQL):** Load data into a relational database and execute targeted business queries addressing location performance, payment trends, seasonal demand, and customer loyalty.
3. **Visualization & Insights (Power BI):** Build an interactive dashboard featuring dynamic KPIs and visual breakdowns of revenue, categories, and customer segments.
4. **Report & Presentation:** Document key findings and actionable business recommendations for stakeholders.
5. **GitHub Repository:** Maintain an organized version-controlled repository containing all Python notebooks, SQL scripts, and Power BI files